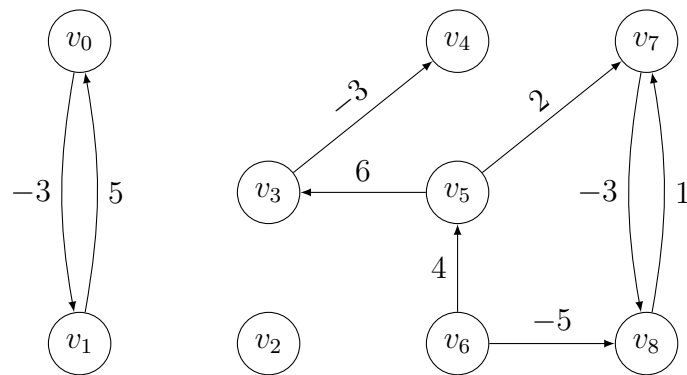


Networks: supply and demand

Michel Bierlaire

Practice quiz

1. Consider the following network and the corresponding vector of flows:



- (a) Calculate the divergence of each node. Which nodes are *supply nodes*? Which ones are *demand nodes*? Which ones are *transit nodes*?
- (b) Verify that the sum of all divergences is zero.
2. Prove that given a network $(\mathcal{N}, \mathcal{A})$ and its associated vector of divergences, the sum of all divergences is always zero.